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Studierichting: Informatica
Oefeningenreeks 2D nr. 14.13

Opgave: Bereken de limiet $\lim_{x\to 0} \frac{2(1-\cos x)}{x^2}$.

Oplossing:

$$\begin{aligned}
\lim_{x\to 0} \frac{2(1-\cos x)}{x^2} &\overset{x=2y}{=} \lim_{y\to 0} \frac{2(1-\cos 2y)}{4y^2} \\
&= \lim_{y\to 0} \frac{4\dfrac{1-\cos 2y}{2}}{4y^2} \\
&= \lim_{y\to 0} \frac{4\sin^2 y}{4y^2} \\
&= \lim_{y\to 0} \frac{\sin^2 y}{y^2} \\
&= \left(\lim_{y\to 0} \frac{\sin y}{y}\right)^2 \\
&= (1)^2 \\
&= 1
\end{aligned}$$

References